MCS4302 Week 4 Lab — Measurement Sheet

Index number: 24441155 **Repository URL:** https://github.com/ThiwaWijethunga/mcs4302-mobile-client
Seeds used: 42, 43, 44 (one complete run per seed; each condition resets to the run seed). 50 sequential logical requests to /flaky per condition; retry = 3 retries, 200 ms base, ±20% jitter.

Averages across three runs

Condition	Success %	Mean ms	Mean of run p95 (ms)	HTTP calls	Theory %
A: 0.3, no retry	66.7	52	80	50.0	70.00
B: 0.3, retry	98.7	209	888	73.7	99.19
C: 0.6, no retry	40.0	52	80	50.0	40.00
D: 0.6, retry	90.7	546	1655	111.0	87.04

Latency spans first attempt to final outcome (including backoff and failures). p95 is nearest rank (48th of 50) per run, then averaged; HTTP calls count /flaky attempts only.

Per-run results (CSV from each run)

Condition	Seed	Success %	Mean ms	p95 ms	HTTP calls
A: 0.3, no retry	42	62.0	55	82	50
B: 0.3, retry	42	98.0	217	793	74
C: 0.6, no retry	42	32.0	53	81	50
D: 0.6, retry	42	88.0	611	1724	115
A: 0.3, no retry	43	76.0	51	80	50
B: 0.3, retry	43	98.0	174	357	71
C: 0.6, no retry	43	50.0	53	80	50
D: 0.6, retry	43	92.0	407	1543	100
A: 0.3, no retry	44	62.0	50	79	50
B: 0.3, retry	44	100.0	235	1512	76
C: 0.6, no retry	44	38.0	50	80	50
D: 0.6, retry	44	92.0	619	1697	118

Idempotency test (seed 42, 10 logical creates, 50% failure injected after save)

Test	Requested	Created	Duplicates	Reported failed
Key OFF	10	19	9	0
Key ON	10	10	0	0

Without a key: 9 duplicates; with a key: 0 duplicates. Reported failures: 0 (key OFF), 0 (key ON) — every injected failure was absorbed by a retry, but without a key each retry re-saved the item.

Interpretation

Closeness to theory. Measured success rates (66.7%, 98.7%, 40.0%, 90.7%) are close to the expected 70%, 99.19%, 40% and 87.04%; the gaps are within normal sampling variation for 150 requests per condition.

Benefit versus cost. At f = 0.3, retry raised success by 32 percentage points for 47% more HTTP calls, but mean latency rose from 52 to 209 ms and p95 from 80 to 888 ms, because the slow tail is made of requests waiting through backoff. At f = 0.6 the gain was larger (+51 points) but cost 2.2× the calls and a p95 of 1655 ms. Retrying more at f = 0.6 has diminishing returns (4 retries ≈ 92.2%, 5 ≈ 95.3% in theory) while each extra retry adds 1.6–3.2 s of backoff and more load on a server that is already failing, so a failure that high is better handled by fixing the server or a circuit breaker.

Jitter. Without jitter, 10,000 phones that failed together would all retry at exactly 200, 400 and 800 ms, producing synchronized spikes (a thundering herd) that can keep an overloaded server failing; ±20% jitter spreads those retries out.